CLINICAL MACHINE VISION

Automated Facial Video Recognition of Depression & Anxiety Severity

Cross-Corpus Validation of Dynamic Facial Biomarkers

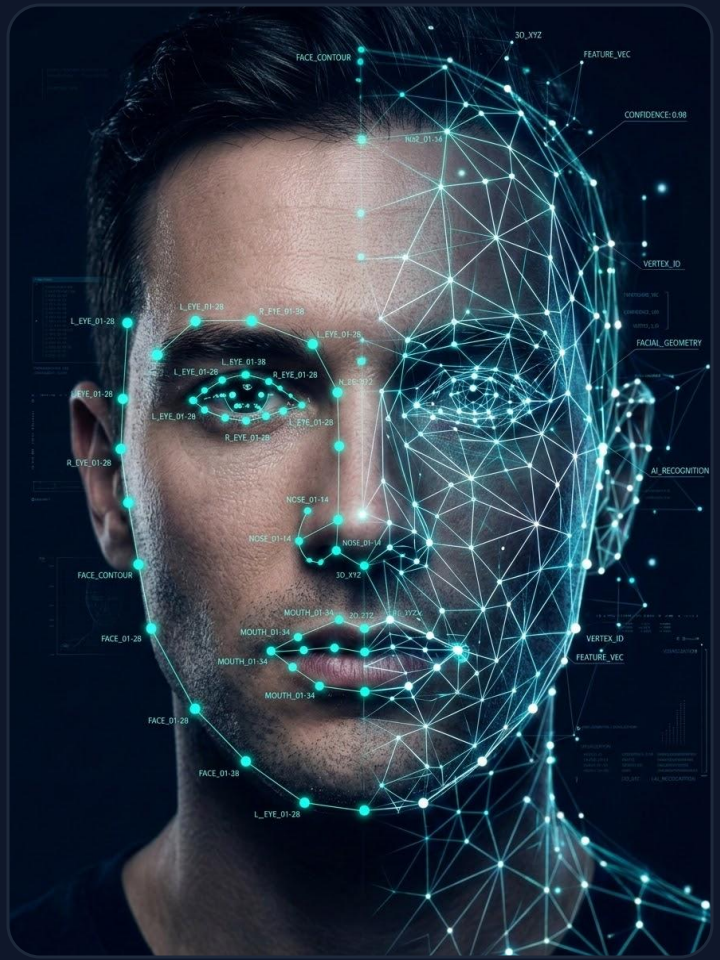
Machine Vision and Applications (Springer, 2020) | DOI: 10.1007/s00138-020-01080-7

Pampouchidou | Pediaditis | Kazantzaki | et al.

Univ. of Burgundy | FORTH-ICS | Univ. of Crete Psychiatry

Research Objective

Establishing an automated, non-invasive computational framework for continuous symptom monitoring across clinical and benchmark datasets.

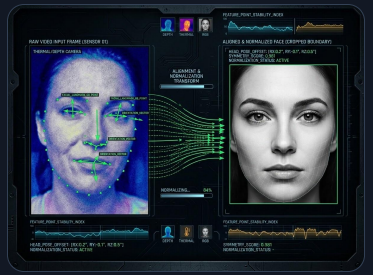


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Proposed Methodology: 4-Stage Computational Pipeline

1. Face Preprocessing

- **OpenFace 2.0 CE-CLM:** landmark detection & facial tracking
- **3D Alignment:** rigid alignment & head-pose normalization



2. Motion Encoding

- **MHI Rep:** Motion History Image converts temporal video dynamics into static images

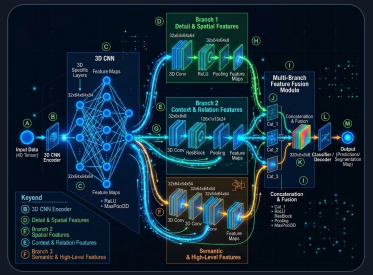
$$H_D(x,y,t) = \max(A, H_D - d)$$



3. Multi-Modal Extraction

HANDCRAFTED
LBP + HOG descriptors

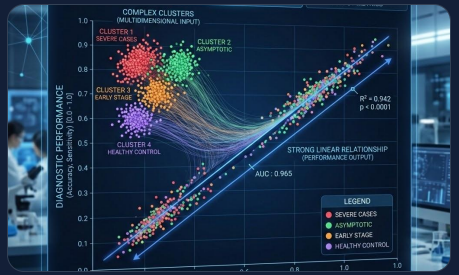
DEEP FEATURES
VGG-16 / VGG-19 CNNs



4. Modeling & SVR

- **PCA Reduction:** mapping high-dim features to 5–50 key components
- **Linear SVR:** Continuous severity regression mapping
- **Linear SVM:** Robust classification boundary

VALIDATION: LOSO PROTOCOL



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Datasets & Experimental Protocols

CLINICAL SESSION TIMELINE (60–90 MIN PROTOCOL)

Baseline Relaxation ▶ Positive Recall / Joy Clip ▶ Breathing Pause ▶ Neutral Reading ▶ Negative Recall / Sad Clip ▶ Questionnaires

Clinical Cohort (Univ. Hospital of Heraklion)

COHORT
65 Subjects
(20 MDD Patients vs. 45 Healthy Controls)

HARDWARE
1920×1920 px @ 80 FPS
(Grasshopper 3 Camera, Indirect Lighting)

PROTOCOL
Social interviews (joy/sadness recall) + non-social video watching + neutral reading

ANNOTATIONS
BDI-II (Depression), **STAI-Trait** (Anxiety), Blinded Expert Video Ratings (**CARMA**, ICC=0.85)

Benchmark Corpus (AVEC'14 Germany)

COHORT
83 Subjects
(300 video recordings)

HARDWARE
640×480 px @ 30 FPS
(Standard Webcam)

PROTOCOL
"Northwind" standardized text reading + freeform dialogue

ANNOTATIONS
Self-reported continuous **BDI-II scores**

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Key Results: Continuous Assessment Performance

Continuous regression achieved high fidelity across both datasets, significantly outperforming binary classification (F1 < 56.4%).

STAI Trait Anxiety (HOG / Neutral Read)

RMSE 9.94 | Normalized Error: 12.42%

BDI-II Depression (VGG-19 FC7 / Neutral Read)

RMSE 10.54 | Normalized Error: 16.73%

AVEC'14 Generalization (HOG / Northwind Task)

RMSE 11.45 | Matches Published SOTA Benchmarks

Binary Classification (High vs. Low Severity)

F1-Score: 56.4% | Baseline Performance

Bland-Altman Agreement



Minimal systematic drift with **only 2** false negatives on high clinical severity.

Key Takeaway

Neutral passage reading yielded the lowest RMSE across both depression and anxiety targets, serving as the most stable computational signal for continuous severity mapping.

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Major Findings & Clinical Insights

CLINICAL VALIDATION

Continuous Spectrum vs. Binary Labels

Depressive symptoms manifest on an **unbroken dimensional spectrum**. Continuous regression (SVR) tracked symptom severity accurately, whereas binary classification failed due to symptom overlap and remission states.

SPECIFICITY

Pronounced Nonverbal Distress

The facial vision pipeline showed **higher sensitivity** to self-reported trait anxiety (STAI) than depression (BDI-II), highlighting that nonverbal facial tension is dominated by anxiety and distress components.

TASK DESIGN

Reading Aloud Outperforms Elicitation

Standardized passage reading produced the **lowest prediction errors**, avoiding the emotional volatility, self-consciousness, and erratic movements caused by evocative tasks.

TELEHEALTH FEASIBILITY

Standard Webcams Are Sufficient

High-resolution video (80 fps) offered **no significant predictive gain** over standard webcams (30 fps), demonstrating high viability for remote, low-cost patient monitoring.